# Review Questions

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## 1. How many distortion coefficients are there and what are they?

The standard OpenCV camera distortion model uses five coefficients: k1, k2, p1, p2 and k3. The k coefficients correct radial distortion, which makes objects appear stretched or squeezed. The p coefficients correct tangential distortion caused by misalignment between the lens and image sensor. OpenCV supports additional coefficients for more complex distortion models.

## 2. Difference between .npz and .yaml files

.npz stores NumPy arrays in an archive and is commonly loaded using np.load(). .yaml stores structured, human-readable data and can be loaded using cv2.FileStorage(). Both can store camera calibration values, such as the camera matrix and distortion coefficients. The choice of format does not affect calibration accuracy.

## 3. What does cv2.moments() do? Explain its attributes.

cv2.moments() calculates mathematical properties of an image region or contour, helping determine its area, centre and shape.

- m00: Area of the region.
- m10, m01: First-order moments used to calculate the centroid.
- m20, m11, m02: Second-order moments describing the distribution of the shape.
- m30, m21, m12, m03: Third-order moments describing further details of its distribution.
- mu20, mu11, mu02, mu30, mu21, mu12, mu03: Central moments measured relative to the centroid.
- nu20, nu11, nu02, nu30, nu21, nu12, nu03: Normalized central moments, useful for comparing shapes of different sizes.

In my project, m00, m10 and m01 help calculate an object's centre for tracking. If m00 is zero, the centroid cannot be calculated using these formulas.

## 4. Difference between Gaussian blur and morphologyEx()

Gaussian blur smooths an image by averaging nearby pixel values with different weights. It reduces noise but can soften object edges. For example, cv2.GaussianBlur(frame, (5, 5), 0) applies Gaussian smoothing.

cv2.morphologyEx() performs shape-based operations, usually on binary masks. It uses a kernel called a structuring element.

- Erosion: Shrinks foreground regions and removes small spots.
- Dilation: Expands foreground regions and can connect nearby areas.
- Opening: Erosion followed by dilation; removes small foreground noise.
- Closing: Dilation followed by erosion; fills small holes and gaps.

In my tracker, morphology cleans the HSV masks before contour detection. Gaussian blur reduces image noise, while morphology improves the shapes of detected regions. They serve different purposes and can be used together.

## 5. Difference between Original ArUco and the 5×5 dictionary

Both use a 5×5 internal binary grid, but their marker patterns and ID systems differ. Original ArUco contains 1,024 markers, while the OpenCV 5×5 dictionaries contain 50, 100, 250 or 1,000 markers, depending on the selected dictionary. A marker must be detected using the same dictionary that was used to generate it. In my project, I need to select the correct dictionary for the tags attached to my objects.

## 6. What functions are available in cv2.aruco?

Common functions and classes include:

- getPredefinedDictionary(): Loads a marker dictionary.
- generateImageMarker(): Generates a marker image.
- ArucoDetector(): Creates a marker detector.
- DetectorParameters(): Sets detection parameters.
- detectMarkers(): Detects marker corners and IDs.
- drawDetectedMarkers(): Draws detected markers and their IDs.
- estimatePoseSingleMarkers(): Estimates individual marker poses in OpenCV versions that provide this API.
- GridBoard() and CharucoBoard(): Create marker-board layouts.
- interpolateCornersCharuco(): Finds ChArUco chessboard corners.
- estimatePoseBoard(): Estimates a supported marker board's pose.

The available API depends on the installed OpenCV version. My project uses marker detection and cv2.solvePnP() to estimate distance and orientation.

## 7. Difference between Hough circles and circularity. Which is better?

cv2.HoughCircles() detects circular patterns in an image and estimates their centres and radii. It can work without first extracting object contours, but it requires suitable parameters and may produce false detections.

Circularity measures how closely an already detected contour resembles a circle:

$$\text{Circularity} = 4\pi \times \text{Area} / \text{Perimeter}^2$$

A perfect circle has a circularity close to 1. Hough circles find circles; circularity evaluates the roundness of existing shapes.

For my tennis-ball tracker, circularity is useful because HSV segmentation already isolates likely balls. Hough circles may be better when detecting circles directly from grayscale images. The best method depends on image quality and the detection requirements.

## 8. Difference between methods and functions

A function is a reusable block of code called by its name, such as cv2.cvtColor(). A method is associated with an object or class and is called through that object, such as cap.read() or detector.detectMarkers(). Both perform operations and may accept inputs or return results.

## 9. What is cv2.Rodrigues()?

cv2.Rodrigues() converts a rotation vector into a 3×3 rotation matrix, or vice versa. In ArUco tracking, cv2.solvePnP() returns a rotation vector (rvec) and translation vector (tvec). Rodrigues conversion makes the orientation easier to use in further calculations. The translation vector describes position relative to the camera.

## 10. Why use HSV instead of BGR?

BGR represents colours using blue, green and red intensity channels. HSV separates colour into hue, saturation and brightness (value), making colour-based segmentation easier.

In my project, I convert frames from BGR to HSV and use cv2.inRange() to isolate blue objects and tennis balls. HSV makes colour thresholds more convenient under moderate lighting changes, although shadows, reflections and very low brightness can still affect detection. Morphological operations then help clean the resulting masks.